

Digital art (2D/3D)

Level designers can never work wonders if the digital art is not up to the mark. In production phase after the concept art is full-proof and accepted by the executives, the 3D/2D artists can start their work. They create the required graphics, illustrations, 3D models, terrains, and landscapes etc. which form the objects seen in the game levels and provide these assets to level designers. These objects can be roughly categorized into static objects, dynamic objects and special effects.

Mesh and polygon modeling

If you ever open a 3D modeling tool, the first thing you can do with it is to create a polygon primitive: cube, sphere, cylinder, etc. These are instant 3D objects. You can understand that these objects if you understand polygons from your basic mathematics background. These polygons are used to create polygon meshes used to create 3d objects and models. The polygon faces usually considered are triangles, squares etc rather than complex polygons for ease of rendering. There are two styles of polygon modeling:

- ▶ **Box modeling:** Box modeling is so named because it starts with a box (a cube primitive). The artist then scales the object, moves (translates) vertices and edges, and adds more polygon faces by adding edges, so that they create the 3D shape which they want.
- ▶ **Extrusion (edge) modeling:** Little different from pulling and reshaping, it's a way to add more polygons to a mesh. The artist can start with an edge or a polygon. The artist selects either an edge or a polygon face and pulls it to add more polygons to the mesh thus creating a 3D form around its boundary. Usually, the artist creates a 2d shape (polygon) which traces the outline of an object from a photograph or drawing. Then he uses the same image and extrudes the 2D shape into 3D.

Mirroring

Instead of modeling each side of the object, only one side is modeled. Then its mirror is duplicated and merged into the mesh. Some applications allow you to see the mirror image as you model. This saves a lot of time and ensures a symmetrical model.

Creating edge loops

You can add additional detail in your model by adding edges in strategic places. Some applications allow you to add a ring of edges that loops around